

TrES-1b

R-0190

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_240821 · created 2026-07-16T20:59:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460543.8114 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.053
Transit depth (Rp/Rs)^2	3.32 +/- 1.93 [%]
Orbital Inclination (inc)	88.24 +/- 1.46
Airmass coefficient 1 (a1)	0.91 +/- 0.53
Airmass coefficient 2 (a2)	0.0 +/- 0.24
Scatter in the residuals of the lightcurve fit is	57.7 %
Best Comparison Star	#1 - [255, 198]
Optimal Aperture	25.86
Optimal Annulus	152.1
Transit Duration (day)	0.1 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.053 vs 0.1358 (deviation 0.62 σ)
Duration fit vs published	0.1 d vs 0.1042 d (deviation 0.27 σ)
Mid-time O-C	-7.5 min
Depth SNR	2.5
Residual scatter	57.7 %

Light curve

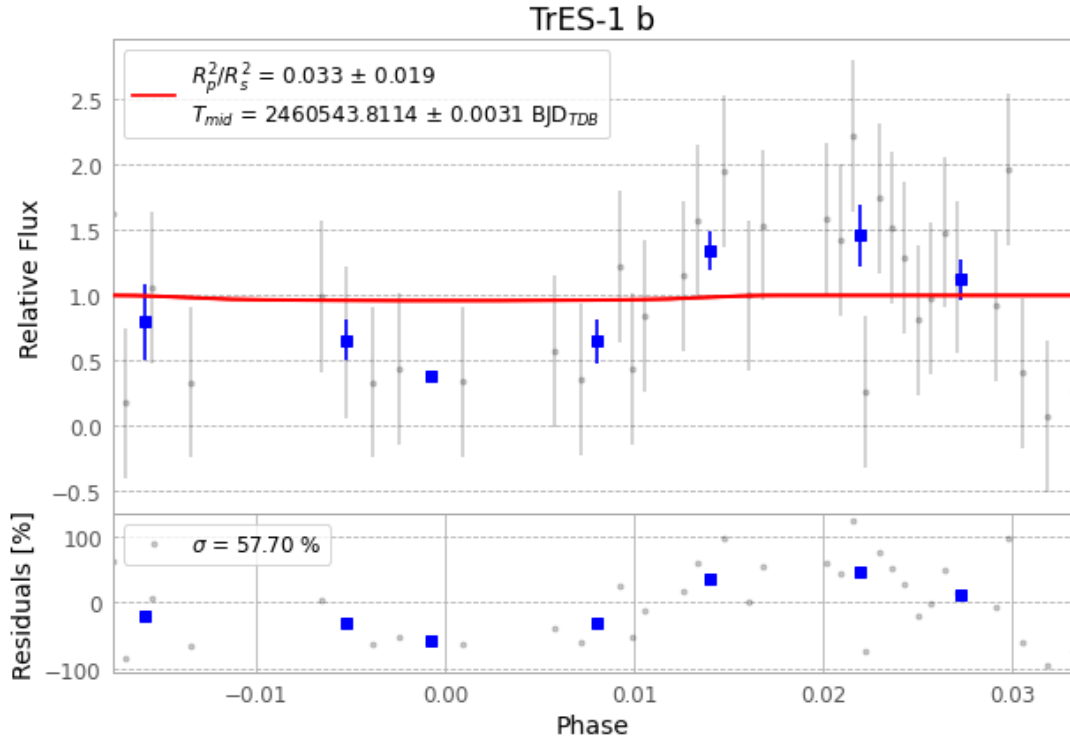


Figure 1 — FinalLightCurve_TrES-1 b_2024-08-20.png (SHA-256 469d83dbc073ab8f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a35f945ebf85ba21...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d

Data provenance

75 FITS frames (49 MB), TRES-1240821060615.FITS → TRES-1240821094816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-240821.log (fbc01ef1444d...), FinalLightCurve_TrES-1 b_2024-08-20.png (469d83dbc073...), FinalLightCurve_TrES-1 b_2024-08-20.csv (9b6e8d5927b9...)

Compute resources

Wall time 1106.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 20:59Z.